

PROJECT REPORT

On

“Samsung Supply Chain & Logistics Dashboard”

ಡಾ. ಮನಮೋಹನ್ ಸಿಂಗ್
ಬೆಂಗಳೂರು ನಗರ ವಿಶ್ವವಿದ್ಯಾನಿಲಯ



DR. MANMOHAN SINGH
BENGALURU CITY UNIVERSITY

Submitted in partial fulfilment of the Sessional Requirements for
6th semester “BACHELOR OF COMPUTER APPLICATIONS”

By

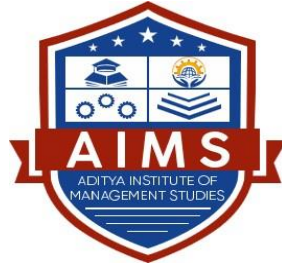
Khushboo Maurya

under the guidance of

Mrs. Yeshodha. R

Assistant professor, Department of Computer Applications

ADITYA INSTITUTE OF MANAGEMENT STUDIES



#12, Kogilu Main Road, Yelahanka, Bangalore-560064

(Affiliated to Bangalore City University, Bangalore)

<http://aimsbangalore.net/> - Phone: 097422 92012

ACKNOWLEDGMENT

We are indebted to our cherished Principal, **Prof. Dr. PUSHPA SHETTY, Director of AIMS** whose support and permission were instrumental in undertaking this dissertation. Their guidance throughout this challenging endeavour has been invaluable. We are deeply grateful to our respected Head of the Department, whose support and permission were instrumental in enabling us to embark on this dissertation journey and guiding us through its completion. Our sincere appreciation goes to our dedicated guide

R. Sangeetha, for their invaluable mentorship and timely assistance in bringing our project to fruition. We are also indebted to the members of the Computer Department for their guidance and support throughout this endeavour. Last but not least, we express our heartfelt gratitude to our beloved parents and dear friends for their unwavering encouragement and inspiration.

Khushboo maurya

Abstract

In today's rapidly evolving business environment, efficient supply chain and logistics management is essential for optimizing operations, reducing costs, and improving decision-making. This project presents the design and development of a Supply Chain & Logistics Dashboard for Samsung, developed using Power BI along with Python and Pandas for advanced data processing and analysis. The system is designed to manage and analyze key supply chain functions such as supplier performance, production efficiency, inventory management, shipment tracking, and sales performance.

The core strength of the dashboard lies in its interactive, scalable, and user-friendly design, supported by data integration and analytical capabilities. Python and Pandas were used for data cleaning, transformation, and creation of calculated metrics such as profit margin and delay analysis. The dashboard consists of multiple modules including Executive Overview, Supplier Analysis, Production Overview, Inventory Management, Shipment Tracking, and Sales Insights, each providing meaningful visualizations and key performance indicators.

The system enables different stakeholders to access relevant insights efficiently. Executives can monitor overall performance, suppliers can be evaluated based on procurement metrics, production teams can track defect rates and output, and inventory managers can maintain optimal stock levels. Shipment analysis helps identify delays and logistics performance, while sales insights support revenue and profitability analysis. This solution provides an effective alternative to traditional reporting methods by offering real-time insights and data-driven decision-making. It reduces manual effort, improves accuracy, and enhances coordination across the entire supply chain.

Future enhancements may include real-time data integration using APIs, AI/ML-based predictive analytics, mobile compatibility, and advanced interactive features. As organizations increasingly rely on data-driven strategies, such dashboards play a vital role in improving efficiency, transparency, and strategic planning.

This project demonstrates the practical application of data analytics, visualization, and Python integration in supply chain management, highlighting the importance of intelligent dashboards in modern business environments.

TABLE OF CONTENT

Serial No.	TITLE	Page NO
1	Introduction	
1.1	Aim of the project	
1.2	Scope & Objective	
1.3	Key components & Data Setup	
1.4	Features & Enhancement of Dashboard	
2	Problem & Solution in Supply Chain Analysis	
2.1	Problems in Traditional Data Analysis	
2.2	Solutions using Power BI Dashboard	
3	System analysis	
3.1	Data Collection & Requirement Analysis	
3.2	Dashboard Architecture	
3.3	Potential Challenges in Data Analysis	
3.4	Implementation plan	
3.5	Testing & Evaluation of Dashboard	
3.6	Deployment	
3.7	Maintenance & optimization	
3.8	Data Security & Compliance	

4	System Requirement	
4.1	Hardware requirements	
4.2	Software requirements	
4.3	Additional considerations	
5	System Design	
5.1	Flow Chart	
5.2	Data flow diagram	
5.3	Data Model / ER Diagram	
6	Testing & Validation	
6.1	Dashboard Testing	
6.2	Data Validation & Accuracy Check	
6.3	Database	
7	Dashboard Components	
8	Output	
9	Conclusion	
10	Bibliography	

1.INTRODUCTION

1. INTRODUCTION

A Supply Chain & Logistics Dashboard is a digital solution designed to streamline and monitor the end-to-end operations of supply chain management. With the increasing need for efficiency, accuracy, and real-time insights in business operations, such dashboards play a crucial role in reducing manual effort, minimizing errors, and improving overall decision-making. This project focuses on developing an interactive Supply Chain & Logistics Dashboard for Samsung using Power BI, integrated with Python and Pandas for advanced data processing and analysis.

The dashboard is structured into multiple functional modules, including Executive Overview, Supplier Analysis, Production Overview, Inventory Management, Shipment Tracking, and Sales Insights. Each module provides specific insights and key performance indicators (KPIs) such as revenue, profit, order quantity, stock levels, defect rates, and delivery performance. These features enable users to easily track and evaluate different stages of the supply chain.

Python integration enhances the system by enabling efficient data cleaning, transformation, and creation of calculated metrics such as profit margin and shipment delay indicators using the Pandas library. This ensures accurate data analysis and supports deeper insights beyond standard visualization tools.

By digitizing and visualizing supply chain processes, the dashboard reduces dependence on traditional reporting methods and provides a centralized platform for monitoring operations. It improves transparency, coordination, and efficiency across various business functions such as procurement, production, inventory, logistics, and sales.

Furthermore, the system supports data-driven decision-making and offers scalability for future enhancements. Advanced features such as real-time data integration, predictive analytics, and interactive filtering can be incorporated to further improve performance and usability. As businesses increasingly adopt digital transformation, dashboards like this play a vital role in enabling smarter, faster, and more efficient supply chain management.

1.1 AIM OF THE PROJECT

The aim of this project is to design and develop an efficient, interactive, and user-friendly Supply Chain & Logistics Dashboard that streamlines business operations and supports data-driven decision-making. The project utilizes Power BI along with Python and Pandas to analyze and visualize key supply chain activities such as supplier performance, production efficiency, inventory management, shipment tracking, and sales analysis. It aims to reduce manual effort, improve data accuracy, and provide meaningful insights through interactive dashboards. By the end of the project, the goal is to deliver a scalable and functional dashboard that demonstrates the importance of data analytics and visualization in optimizing supply chain performance.

1.2 SCOPE & OBJECTIVE

Scope:

- **Executive Overview:** Provide a high-level summary of key performance indicators such as revenue, profit, order quantity, and overall business performance.
- **Supplier Analysis:** Evaluate supplier performance based on procurement cost, lead time, and category-wise contribution.
- **Production Monitoring:** Track production output, defect rates, and efficiency to improve manufacturing performance.
- **Inventory Management:** Monitor stock levels, identify understock and overstock situations, and maintain optimal inventory.
- **Shipment Tracking:** Analyze shipment performance, delivery status, delays, and logistics costs.
- **Sales Insights:** Provide detailed analysis of revenue, profit, customer performance, and product profitability.
- **Data Processing using Python:** Use Python and Pandas for data cleaning, transformation, and creation of calculated metrics such as profit margin and delay indicators.
- **Data Visualization:** Utilize Power BI to create interactive and user-friendly dashboards for better understanding of data.

- **Decision Support:** Enable users to make informed decisions based on real-time insights and trends.
- **Scalability:** Design the dashboard to support future enhancements such as real-time data integration, predictive analytics, and advanced filtering options.

Objectives:

- **Efficient Data Management:** Streamline the process of collecting, cleaning, and managing supply chain data to ensure accuracy and consistency.
- **Interactive Dashboard:** Develop a user-friendly and interactive dashboard using Power BI for easy navigation and data visualization.
- **Supplier Performance Analysis:** Evaluate suppliers based on procurement cost, lead time, and overall contribution to improve decision-making.
- **Production Monitoring:** Track production output, defect rates, and efficiency to enhance manufacturing performance.
- **Inventory Optimization:** Monitor stock levels to identify understock and overstock situations and maintain optimal inventory levels.
- **Shipment Analysis:** Analyze shipment status, delivery performance, and delays to improve logistics efficiency.
- **Sales and Profit Analysis:** Provide insights into revenue, profit, and product performance to support business growth.
- **Python Integration:** Utilize Python and Pandas for data cleaning, transformation, and creation of calculated metrics such as profit margin and delay indicators.
- **Data Visualization:** Present data through meaningful charts and KPIs to support quick and informed decision-making.
- **Scalability and Enhancement:** Design the system to allow future improvements such as real-time data integration and predictive analytics.

With these objectives, the project effectively addresses key supply chain operations, improving efficiency, transparency, and overall business performance.

1.3 KEY COMPONENTS & DATA SETUP

Key Components:

1. Requirement Analysis and Planning

- **Dashboard Planning:** Identify key modules such as Executive Overview, Supplier Analysis, Production, Inventory, Shipment, and Sales.
- **KPI Identification:** Define important metrics like revenue, profit, stock levels, defect rate, and delivery performance.

2. Data Collection and Preparation

- **Data Sources:** Gather supply chain data from CSV/Excel files.
- **Data Cleaning:** Use Python and Pandas to handle missing values, remove errors, and format the dataset properly.

3. Data Processing using Python

- **Transformation:** Create calculated fields such as profit margin and delay indicators.
- **Analysis:** Perform data manipulation to prepare meaningful datasets for visualization.

4. Dashboard Development (Power BI)

- **Tool Used:** Power BI for creating interactive and dynamic dashboards.
- **Visualization:** Design charts, graphs, and KPI cards for better understanding of data.

5. Data Modeling

- **Relationships:** Establish relationships between different data tables.
- **Optimization:** Ensure efficient data flow and fast performance within the dashboard.

6. Integration and Testing

- **Data Integration:** Load processed data into Power BI.

- **Testing:** Verify accuracy of visuals, KPIs, and data consistency across all modules.

7. Deployment and Usage

- **Dashboard Sharing:** Publish dashboard for users to access insights.
- **User Interaction:** Enable filters, slicers, and drill-down features for better analysis.

8. Maintenance and Enhancement

- **Updates:** Refresh data regularly to keep insights up-to-date.
- **Future Improvements:** Add features like real-time data integration and predictive analytics.

These components ensure the successful development of a reliable, efficient, and user-friendly Supply Chain & Logistics Dashboard.

Setup:

1. Hardware and Software Setup

- **Hardware Requirements:** A system with adequate processing power and memory to run Power BI and Python efficiently.
- **Software Tools:**
- **Visualization Tool:** Power BI Desktop
 - **Programming Language:** Python
 - **Libraries:** Pandas, Matplotlib
 - **Data Source:** CSV/Excel files
 - **Environment:** Python installed and configured with Power BI

2. Data Infrastructure

- **Data Storage:** Supply chain data stored in CSV or Excel format for easy access and processing.

- **Data Processing:** Python and Pandas used for data cleaning, transformation, and preparation before loading into Power BI.

3. Development Environment

- **Tools Used:** Power BI Desktop for dashboard creation and Visual Studio Code or similar IDE for Python scripting.
- **Version Control:** Basic file management used to maintain dataset and dashboard versions.

4. Data Processing and Integration

- **Python Integration:** Python scripts are used within Power BI to load and process data.
- **Data Transformation:** Creation of calculated fields such as profit margin and delay indicators using Pandas.

5. Testing and Deployment

- **Testing:** Validate data accuracy, KPIs, and dashboard visuals within Power BI.
- **Deployment:** Dashboard can be shared using Power BI Service or exported for presentation.

This setup ensures smooth development and efficient functioning of the Supply Chain & Logistics Dashboard.

1.4 Features & Enhancements of Dashboard

- **Modular Dashboard Design:** The dashboard is organized into multiple modules such as Executive Overview, Supplier Analysis, Production, Inventory, Shipment, and Sales, making it scalable and easy to maintain.
- **Optimized Performance:** The system is designed to run efficiently on standard hardware, ensuring smooth performance in Power BI without requiring high-end systems.
- **Efficient Data Handling:** Python and Pandas enable effective data processing even with moderate datasets, ensuring quick analysis and accurate results.

- **Interactive and User-Friendly Interface:** The dashboard provides an intuitive interface with filters, slicers, and drill-down features for easy navigation and better user experience.
- **Advanced Data Processing with Python:** Integration of Python allows data cleaning, transformation, and creation of calculated metrics such as profit margin and shipment delay indicators.
- **Comprehensive Data Visualization:** Power BI is used to create dynamic charts, graphs, and KPI cards that help users easily understand trends and performance.
- **Real-Time Insight Capability (Extendable):** The system can be enhanced to support real-time data integration using APIs for up-to-date analysis.
- **Scalability and Flexibility:** The dashboard is designed to accommodate future enhancements such as predictive analytics and advanced filtering options.
- **Cross-Platform Accessibility:** The dashboard can be accessed on different systems and can be shared through Power BI Service for wider accessibility.
- **Future Enhancements:** The system can be extended with AI/ML-based forecasting, automated alerts, and advanced analytics to further improve decision-making.

These features ensure that the dashboard is efficient, scalable, and capable of supporting modern supply chain analytics.

2. Problem & Solution in Supply Chain Analysis

2.1 Problems in Traditional Data Analysis

Data Quality Issues: Supply chain data may contain missing values, duplicates, or inconsistencies, which can affect the accuracy of analysis and visualizations. Poor data quality leads to incorrect insights and decision-making.

- Data Integration Challenges: Combining data from multiple sources such as CSV or Excel files and maintaining consistency across datasets can be difficult. Differences in formats and structures may create integration issues.
- Python Integration Issues: Setting up Python in Power BI and managing dependencies like Pandas and Matplotlib can cause configuration errors, affecting data loading and processing.
- Performance Limitations: Handling large datasets can slow down dashboard performance, increase refresh time, and affect user experience.
- Visualization Complexity: Designing meaningful and easy-to-understand visuals for multiple modules like inventory, shipment, and sales requires careful planning and selection of appropriate charts.
- Lack of Real-Time Data: The current dashboard mainly uses static data, which limits the ability to monitor real-time supply chain activities and respond quickly to changes.
- User Understanding and Adoption: Users who are not familiar with data analytics tools may find it difficult to interpret dashboards and insights effectively.

2.2 Solutions using Power BI Dashboard

- **Data Cleaning and Preprocessing:** Python and Pandas are used to clean the dataset by removing null values, correcting inconsistencies, and preparing structured data for analysis.
- **Standardized Data Integration:** Proper formatting and structuring of data ensure smooth integration and consistency across all modules of the dashboard.
- **Correct Python Configuration:** Setting up Python environment properly and installing required libraries ensures smooth execution of scripts in Power BI.
- **Performance Optimization Techniques:** Data modeling, filtering, and limiting unnecessary data improve dashboard speed and responsiveness.
- **Effective Visualization Design:** Use of appropriate charts, graphs, and KPI indicators improves clarity and helps users easily understand insights.
- **Future Real-Time Data Integration:** The system can be enhanced by integrating APIs to fetch real-time data for dynamic updates and monitoring.
- **User-Friendly Interface:** Interactive features like slicers, filters, and drill-down options make the dashboard easy to use for both technical and non-technical users.
- **Basic Data Security Measures:** Access control and secure sharing options in Power BI help protect sensitive business information.

3. SYSTEM ANALYSIS

3. SYSTEM ANALYSIS

System analysis for the Supply Chain & Logistics Dashboard involves evaluating both functional and non-functional requirements to ensure the system meets business needs, provides accurate insights, and performs efficiently.

3.1 Data Collection & Requirement Analysis

Functional Requirements:

- **Data Loading and Integration:** The system should allow importing data from sources such as CSV or Excel files into Power BI for analysis.
- **Data Processing using Python:** Python and Pandas should be used for data cleaning, transformation, and creation of calculated metrics like profit margin and delay indicators.
- **Dashboard Modules:** The system should include multiple modules such as Executive Overview, Supplier Analysis, Production, Inventory, Shipment, and Sales.
- **KPI Monitoring:** The dashboard should display key performance indicators like revenue, profit, stock levels, defect rates, shipment performance, and delivery status.
- **Interactive Visualization:** Users should be able to interact with the dashboard using filters, slicers, and drill-down features for better data analysis.
- **Reporting and Insights:** The system should generate meaningful insights to support decision-making in supply chain operations.

Non-functional Requirements:

- **Performance:** The dashboard should load quickly and handle moderate to large datasets efficiently without lag.
- **Scalability:** The system should support future enhancements such as real-time data integration and predictive analytics.
- **Usability:** The dashboard should have a user-friendly interface that is easy to understand for both technical and non-technical users.
- **Reliability:** The system should provide consistent and accurate results with minimal errors during data processing and visualization.
- **Maintainability:** The dashboard should be easy to update, modify, and maintain for future improvements.
- **Security:** Data access and sharing should be controlled to protect sensitive business information.

This system analysis ensures that the dashboard is efficient, reliable, and capable of supporting effective supply chain decision-making.

3.2 Dashboard Architecture

The Dashboard Architecture of the Supply Chain & Logistics Dashboard is designed to ensure smooth data flow, efficient processing, and interactive visualization. It consists of four main layers: Frontend (Power BI Dashboard), Backend/Data Processing, Database & Storage, and Security & Data Handling.

1. Frontend (Power BI Dashboard)

Technologies Used:

- Power BI Desktop
- Built-in visualization tools (charts, graphs, KPI cards)

Features:

- Interactive dashboard with modules such as Executive Overview, Supplier, Production, Inventory, Shipment, and Sales
- User-friendly interface with filters, slicers, and drill-down features
- Dynamic visualizations for tracking KPIs like Revenue, Profit, Shipping Cost, Quantity, and Delivery Performance
- Easy navigation and real-time interaction with data
- Support for custom visuals, including Python-based charts

2. Backend / Data Processing Layer

Technologies Used:

- Python
- Pandas
- Matplotlib

Features:

- Data cleaning and preprocessing using Pandas
- Creation of calculated metrics such as Profit Margin, On-Time Delivery %, and Shipment Status
- Data transformation and formatting for accurate analysis

- Integration with Power BI using Python scripts
- Ability to perform advanced analysis beyond standard Power BI features

3. Database and Data Storage

Technologies Used:

- CSV / Excel files (primary data source)
- Power BI Data Model

Features:

- Structured storage of supply chain data including Suppliers, Production, Inventory, Shipments, and Sales
- Creation of relationships between tables inside Power BI
- Fast data retrieval through optimized data models
- Easy data refresh and update functionality

4. Security and Data Handling

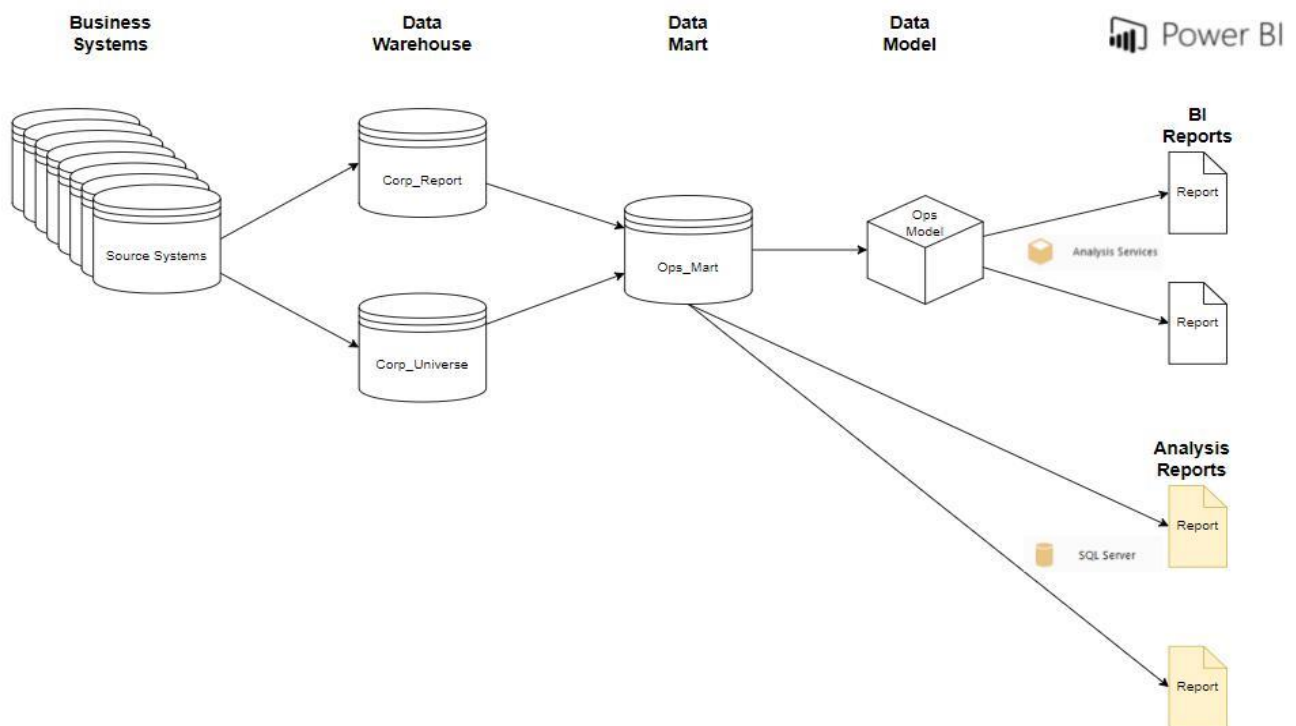
Technologies/Concepts:

- Power BI access control and sharing settings
- Secure local data handling

Features:

- Controlled access to dashboard through permissions
- Data integrity maintained during transformation and visualization
- Secure handling of business and operational data

Dashboard Architecture



3.3 Potential Challenges in Data Analysis

- **Data Accuracy:** Ensuring the dataset is clean, consistent, and free from missing or incorrect values is a major challenge, as inaccurate data can lead to misleading insights and poor decision-making.
- **Scalability:** As the volume of supply chain data increases (more products, suppliers, and transactions), maintaining dashboard performance and efficiency becomes challenging.
- **Real-Time Data Handling:** The current system mainly works on static data, so integrating real-time data updates with minimal delay can be difficult.
- **Performance Optimization:** Large datasets and complex visualizations can slow down dashboard loading and refresh times, affecting user experience.
- **Python Integration Issues:** Configuring Python and managing dependencies like Pandas within Power BI can lead to errors if not set up correctly.
- **Visualization Complexity:** Designing clear and meaningful visuals for multiple modules (inventory, shipment, production, sales) requires careful planning to avoid confusion.
- **User Understanding:** Users with limited knowledge of data analytics tools may find it difficult to interpret insights and use advanced features effectively.
- **Data Security:** Ensuring secure handling and sharing of business data is important, especially when dashboards are shared across users or platforms.

3.4 Implementation Plan

Phase 1: Data Setup

- Collect supply chain data from CSV/Excel files.
- Prepare and organize datasets for analysis.
- Use Python and Pandas to clean data, handle missing values, and format it properly.

Phase 2: Data Processing (Python Integration)

- Perform data transformation using Pandas.
- Create calculated metrics such as profit margin and shipment delay indicators.
- Prepare structured data for visualization in Power BI.

Phase 3: Dashboard Development (Power BI)

- Design dashboard layout with modules like Executive Overview, Supplier, Production, Inventory, Shipment, and Sales.
- Create charts, graphs, and KPI cards for better visualization.
- Implement filters, slicers, and drill-down features for interactivity.

Phase 4: Data Integration

- Load processed data into Power BI using Python scripts or direct data import.
- Establish relationships between data tables for accurate analysis.

Phase 5: Testing and Optimization

- Verify accuracy of data, KPIs, and visuals.
- Optimize dashboard performance by reducing unnecessary data and improving model efficiency.
- Test Python scripts and ensure smooth execution.

Phase 6: Deployment

- Publish the dashboard using Power BI Service or export for presentation.
- Share dashboard with users and enable access controls.
- Monitor performance and update data regularly.

This implementation plan ensures a structured approach to developing an efficient and interactive Supply Chain & Logistics Dashboard.

3.5 Testing & Evaluation of Dashboard

1. Unit Testing

- **Purpose:** To verify individual components such as Python scripts, data transformations, and Power BI visuals.
- **Tools:** Python (Pandas), Power BI validation features.
- **Approach:** Test calculations like profit margin, delay indicators, and data cleaning logic using sample datasets.
- **Outcome:** Ensures each component works correctly before integration.

2. Integration Testing

- **Purpose:** To ensure smooth interaction between Python processing and Power BI dashboard.
- **Scope:** Data loading, transformation, relationships, and visualization across modules like inventory, shipment, and sales.
- **Approach:** Perform end-to-end testing from raw data input to final dashboard output.
- **Outcome:** Confirms proper data flow and accurate insights across the system.

3. Performance Testing

- **Purpose:** To evaluate speed, scalability, and stability of the dashboard.
- **Types:**

- Load Testing: Test performance with large datasets.
- Scalability Testing: Check system behavior as data volume increases.
- Endurance Testing: Run dashboard for long durations to detect slowdowns.
- **Metrics:** Data refresh time, dashboard loading speed, and system responsiveness.

3.6 Deployment

1. **Platform Selection:** Power BI Service is used to publish and share dashboards.
2. **Preparation:**
 - Ensure dataset is clean and structured.
 - Verify Python integration and required libraries.
3. **Configuration:**
 - Set up data connections and scheduled refresh.
 - Validate all visuals, KPIs, and filters.
4. **Post-Deployment:**
 - Share dashboard with users using access permissions.
 - Monitor performance and update data regularly.

3.7 Maintenance and Optimization

1. Maintenance:

- Regular updates of dataset and dashboard components.
- Monitor errors and fix bugs.
- Provide user support and improvements based on feedback.

2. Optimization:

- Improve performance by reducing unnecessary data and optimizing queries.

- Enhance dashboard design for better usability.
- Scale system for handling larger datasets and additional features.

3.8 Data Security & Compliance

1. Data Governance:

- Maintain data consistency, accuracy, and proper handling of business information.

2. Security Measures:

- Use Power BI access control to restrict user permissions.
- Secure data during sharing and publishing.
- Validate and clean data using Python to avoid errors and inconsistencies.

3. Future Security Enhancements:

- Implement advanced access controls and data encryption if integrated with enterprise systems.

This structured testing, deployment, and maintenance approach ensures that the Supply Chain & Logistics Dashboard is reliable, efficient, scalable, and capable of delivering accurate insights for effective decision-making.

4. SYSTEM REQUIREMENT

4. SYSTEM REQUIREMENTS

For the Supply Chain & Logistics Dashboard project, system requirements are defined to ensure efficient development, smooth execution, and reliable performance using Power BI and Python integration.

4.1 Hardware Requirements

Processing Power:

- **CPU:** Minimum Intel Core i5 / AMD Ryzen 5 or higher to handle Power BI processing, large datasets, and Python execution.
- Multi-core processors are preferred for faster data refresh and transformations.

Memory:

- **RAM:** Minimum 8 GB (16 GB recommended) for handling large datasets, running Power BI, and executing Python scripts simultaneously without lag.

Storage:

- **Disk Space:** Minimum 100 GB free space. SSD is recommended for faster data access, loading, and performance optimization.

Display:

- A system with good screen resolution (Full HD or higher) is recommended for better visualization and dashboard design.

4.2 Software Requirements

Operating System:

- Windows 10/11 (64-bit recommended), macOS, or Linux (for Python environment setup).

Tools and Technologies:

- **Power BI Desktop:** Used for building, designing, and publishing dashboards.
- **Python:** Required for advanced data processing and integration.
- **Libraries:**
 - **Pandas** – for data cleaning and transformation
 - **Matplotlib** – for Python-based visualizations
- **Data Sources:** CSV and Excel files containing supply chain data.

Development Environment:

- **IDE/Editors:** Visual Studio Code, Jupyter Notebook, or PyCharm for Python development and testing.

Integration Setup:

- Python should be properly configured in Power BI settings.
- Required libraries (Pandas, Matplotlib) must be installed and accessible.

4.3 Additional Considerations

Data Management:

- Maintain structured and clean datasets for accurate analysis.

- Use consistent formats and naming conventions across all data sources.
- Perform regular data refresh to keep insights up-to-date.

Performance Optimization:

- Use efficient data modeling techniques in Power BI (relationships, star schema).
- Reduce unnecessary columns and rows to improve speed.
- Optimize Python scripts to avoid delays during execution.

Scalability:

- The system should support increasing data volume and additional modules in the future.
- Can be extended with real-time data integration using APIs.

Security:

- Use Power BI sharing permissions to control access.
- Avoid exposing sensitive business data in public dashboards.
- Ensure safe handling of data during import and transformation.

Backup and Recovery:

- Maintain backup copies of datasets and Power BI (.pbix) files.
- Use versioning to track changes and restore previous versions if needed.

Backend (Data Processing using Python)



Python:

Python is used as a backend processing tool to clean, transform, and analyze supply chain data before visualization. It helps automate repetitive tasks and improves data accuracy.



Pandas:

Pandas is used for handling missing values, filtering data, grouping, and creating calculated fields such as profit margin, delivery delays, and performance metrics.



Matplotlib:

Matplotlib is used to generate custom visuals when required, enhancing analytical capabilities within Power BI.

Frontend (Visualization using Power BI)



Power-BI-Dashboard:

Power BI acts as the frontend layer, providing interactive and dynamic dashboards. It allows users to explore data through charts, graphs, KPI cards, slicers, and drill-down features.

User Interaction:

Users can filter data, analyze trends, and gain insights in real-time through an intuitive and visually appealing interface.

Network & System Support:

- Stable internet connection is required for Power BI Service (publishing and sharing dashboards).
- Regular system updates and proper configuration ensure smooth functioning.

By combining Power BI for visualization and Python for data processing, this system ensures high performance, scalability, and efficient supply chain analysis, enabling organizations to make accurate and data-driven decisions.

5. SYSTEM DESIGN

5. SYSTEM DESIGN

System design for the Supply Chain & Logistics Dashboard defines the overall structure, components, data flow, and interactions required to meet the project objectives. It ensures that both functional and non-functional requirements are effectively implemented using Power BI and Python.

Design Overview:

Design is the bridge between data requirements and dashboard implementation. It involves converting raw supply chain data into meaningful insights through structured planning and visualization. The design process includes:

- Preliminary Design:

Identifies key modules such as Executive Overview, Supplier Analysis, Production, Inventory, Shipment, and Sales. It also defines data sources (CSV/Excel) and required KPIs like revenue, profit, stock levels, and delivery performance.

- Detailed Design:

Focuses on data transformation using Python (Pandas), creation of calculated metrics (profit margin, delay indicators), data modeling in Power BI, and designing interactive visual layouts with charts, graphs, and KPI cards.

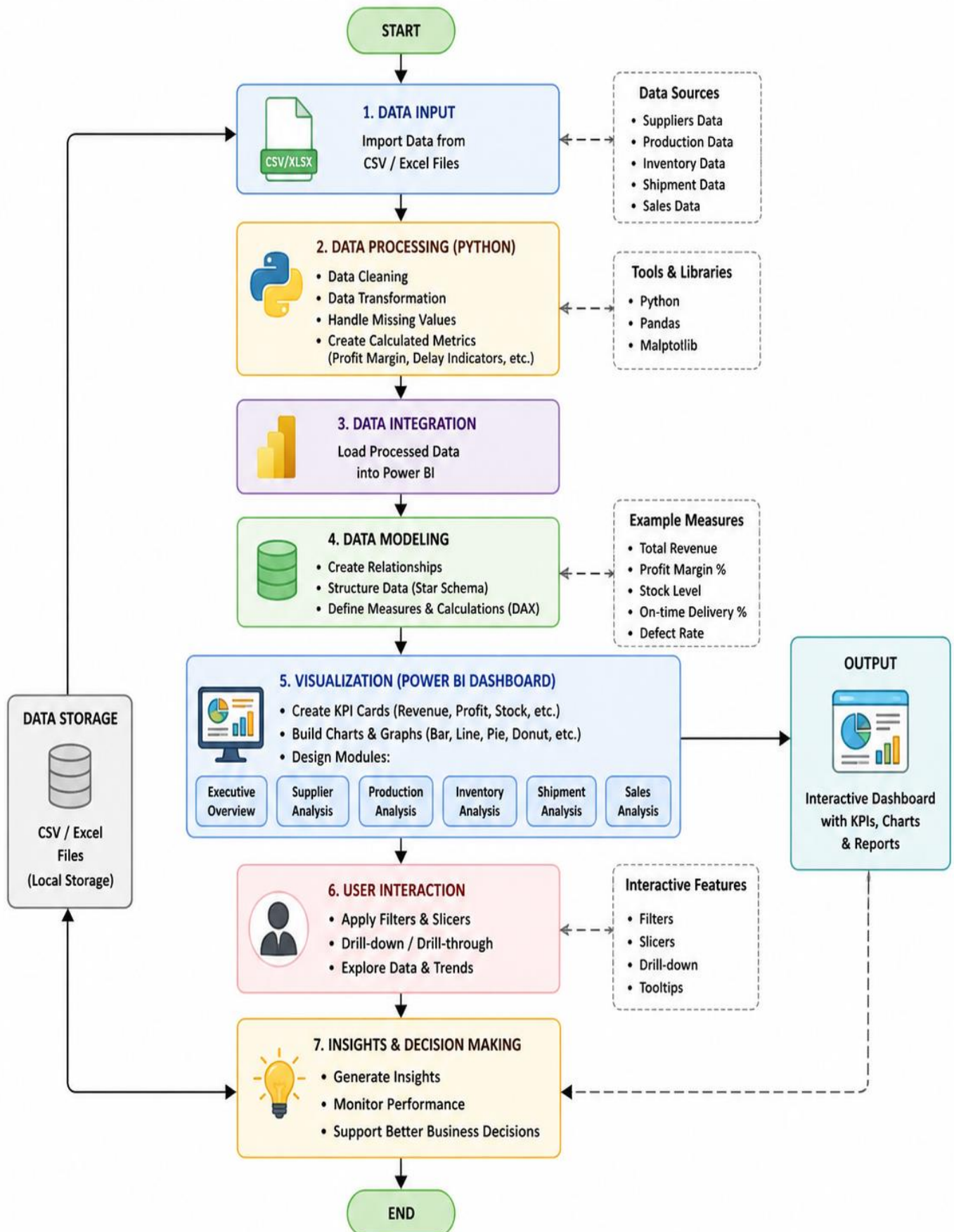
A well-structured design improves dashboard performance, usability, scalability, and maintainability while ensuring accurate and meaningful insights.

5.1 Flow Chart (Dashboard Workflow)

The flowchart represents the complete logical workflow of the Supply Chain & Logistics Dashboard, illustrating how raw data is transformed into meaningful insights for decision-making. It provides a clear visual representation of each stage in the system, starting from data collection to final output, ensuring better understanding, development, and maintenance of the dashboard.

The process begins with **data input**, where raw data is collected from various sources such as CSV or Excel files. This data typically includes information related to suppliers, production, inventory levels, shipment details, and sales performance. Since raw data may contain inconsistencies or missing values, it is then passed to the **data processing stage**, where Python (using Pandas) is used to clean, transform, and structure the data. This step ensures accuracy and prepares the data for analysis.

SUPPLY CHAIN & LOGISTICS DASHBOARD – FLOW CHART



Flowchart Steps Explanation

1. Start

The process begins with initiating the dashboard system.

2. Data Input

- Import raw data from CSV / Excel files
- Data includes: Supplier, Production, Inventory, Shipment, and Sales data

3. Data Processing (Python - Pandas)

- Clean the data (remove errors, duplicates)
- Handle missing values
- Transform data into proper format
- Create calculated fields (Profit %, Delay status, etc.)

4. Data Integration (Power BI)

- Load the processed data into Power BI
- Ensure correct data types and structure

5. Data Modeling

- Create relationships between tables
- Build data model (star schema if needed)
- Define measures using DAX (Revenue, Profit, KPI metrics)

6. Data Visualization (Dashboard Creation)

- Create visuals like:
 - Bar charts
 - Line charts
 - Pie/Donut charts
 - KPI cards
- **Design different modules:**
 - Executive

- Supplier
- Production
- Inventory
- Shipment
- Sales

7. User Interaction

- Apply filters and slicers
- Use drill-down and drill-through features
- Explore trends and comparisons

8. Insights & Decision Making

- Analyze performance
- Identify trends and issues
- Support business decisions

9. End

Final output is an interactive dashboard with insights and reports.

5.2 DATA FLOW DIAGRAM

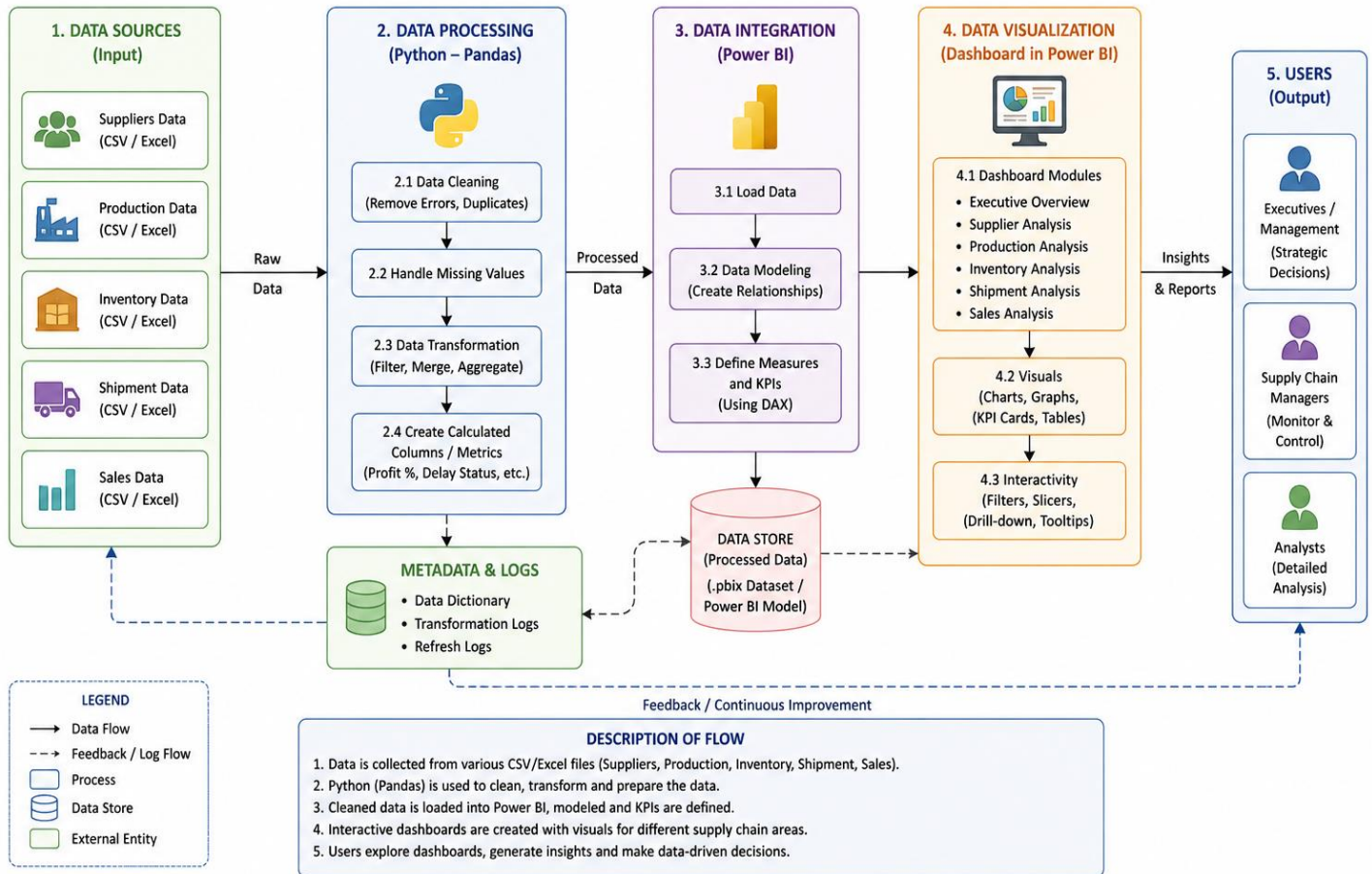
In the context of the Supply Chain & Logistics Dashboard, a Data Flow Diagram (DFD) illustrates how data moves between different components of the system, including data sources, processing tools, and visualization layers. It represents the flow of information from raw data input to meaningful output in the form of dashboards and reports.

The DFD helps in understanding how supply chain data such as supplier details, production records, inventory levels, shipment status, and sales data are collected, processed, stored, and visualized. The process begins with data sources such as CSV or Excel files, which act as the primary input for the system.

This data is then passed to the data processing layer, where Python (using Pandas) is used to clean, filter, and transform the data. Operations such as handling missing values, removing duplicates, and creating calculated metrics (like profit margin, delivery performance, etc.) are performed in this stage.

5.2 DATA FLOW DIAGRAM

Supply Chain & Logistics Dashboard



Supply Chain & Logistics Dashboard

5.3 Data Model / ER Diagram

The Entity-Relationship (ER) Diagram is a conceptual data model used to visually represent the structure of a database by defining entities, their attributes, and the relationships between them. It plays an important role in designing a well-structured and efficient database for the Supply Chain & Logistics Dashboard.

In the context of the Supply Chain & Logistics Dashboard, the ER diagram helps in representing how different data components such as suppliers, products, inventory, shipments, and sales are interconnected within the system. It provides a clear understanding of how data is organized and how different entities interact with each other.

The ER diagram typically includes key entities such as:

- **Supplier** (Supplier_ID, Name, Location, Lead_Time)
- **Product** (Product_ID, Product_Name, Category)
- **Inventory** (Inventory_ID, Stock_Level, Reorder_Point)
- **Shipment** (Shipment_ID, Shipment_Date, Delivery_Status)
- **Sales** (Sales_ID, Revenue, Profit, Customer_Type)

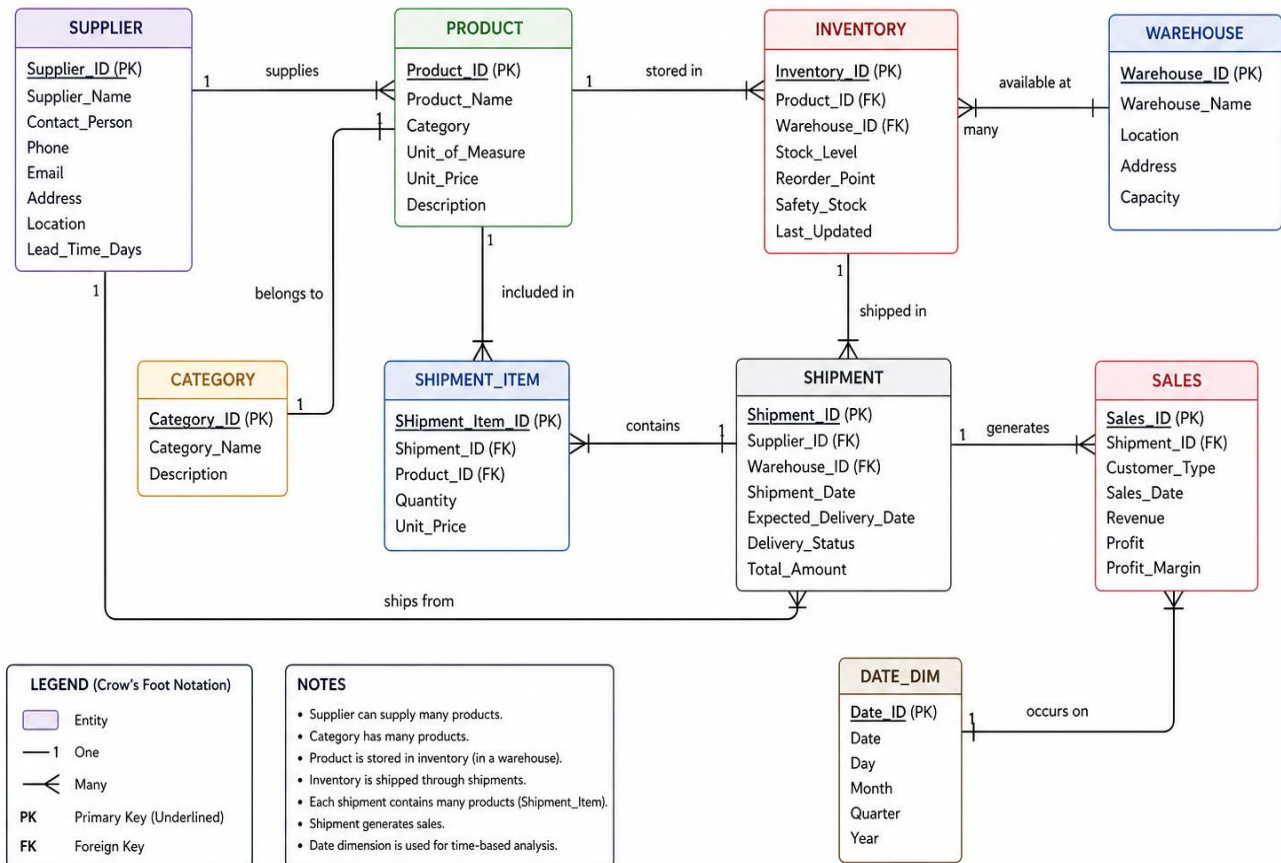
These entities are connected through relationships. For example, a supplier supplies multiple products, products are stored in inventory, inventory is linked to shipments, and shipments contribute to sales data. These relationships help in maintaining data consistency and enable efficient data analysis.

The ER diagram serves the following purposes in this project:

- It provides a clear blueprint of how supply chain data is structured.

- It helps in organizing complex datasets into simple, understandable relationships.
- It supports the creation of relational tables used in Power BI and data processing.
- It ensures data integrity and reduces redundancy in the dataset.

ER DIAGRAM – SUPPLY CHAIN & LOGISTICS DASHBOARD



6. TESTING & VALIDATION

6.1 Dashboard Testing

Testing in the **Supply Chain & Logistics Dashboard using Power BI** ensures that data is accurately loaded, transformed, and visualized. It verifies that all reports, charts, and KPIs provide correct and meaningful insights for decision-making.

1. Data Testing

- **Purpose:** Ensure that the imported data is correct and complete.
- **Steps:**
 - Verify data loaded from Excel/CSV files
 - Check for missing or duplicate values
 - Validate data types (numeric, date, text)
- **Outcome:** Clean and accurate dataset for analysis

2. Data Transformation Testing

- **Purpose:** Validate data cleaning and transformation steps in Power Query / Python
- **Steps:**
 - Check filtering, sorting, and column transformations
 - Verify calculated columns and derived metrics
- **Outcome:** Correctly processed data

3. DAX & KPI Testing

- **Purpose:** Ensure all KPIs and calculations are correct
- **Steps:**
 - Validate formulas for Revenue, Profit, Cost, etc.
 - Cross-check results with manual calculations
- **Example:**

- Profit = Revenue – Cost
- Profit % = (Profit ÷ Revenue) × 100
- **Outcome:** Accurate KPI results

4. Visualization Testing

- **Purpose:** Ensure charts and graphs display correct data
- **Steps:**
 - Verify bar charts, pie charts, KPI cards
 - Check if visuals match dataset values
- **Outcome:** Correct and meaningful visual representation

5. Filter & Interaction Testing

- **Purpose:** Ensure interactivity works properly
- **Steps:**
 - Test slicers (date, channel, region)
 - Check drill-down and cross-filtering
- **Outcome:** Smooth and interactive dashboard

6. Performance Testing

- **Purpose:** Ensure dashboard runs efficiently
- **Steps:**
 - Check loading time of visuals
 - Optimize data model and reduce unnecessary fields
- **Outcome:** Fast and responsive dashboard

6.2 Data Validation & Accuracy Check

Data Validation and Accuracy Check ensure that the Power BI dashboard provides correct, reliable, and meaningful insights. It verifies that the data, calculations, and visuals are accurate and aligned with business requirements.

1. Functional Validation

- **Purpose:** Confirm that all dashboard features work correctly
- **Examples:**
 - Shipment status (Delivered, Delayed, In Transit) is displayed accurately
 - Revenue, Profit, and Shipping Cost calculations are correct
 - Filters and slicers update data properly

2. Data Validation

- **Purpose:** Ensure accuracy and consistency of data
- **Steps:**
 - Compare dashboard results with the original dataset (Excel/CSV)
 - Verify totals, sums, and aggregated values
 - Check for missing, duplicate, or incorrect data

3. Performance Validation

- **Purpose:** Ensure smooth and efficient dashboard performance
- **Examples:**
 - Check report loading time
 - Test responsiveness when applying filters or slicers
- **Outcome:** Fast and responsive dashboard

4. Usability Validation

- **Purpose:** Ensure the dashboard is easy to use and understand
- **Steps:**
 - Verify clear layout and navigation between pages
 - Check readability of charts, labels, and KPIs
- **Outcome:** User-friendly interface

5. Compatibility Validation

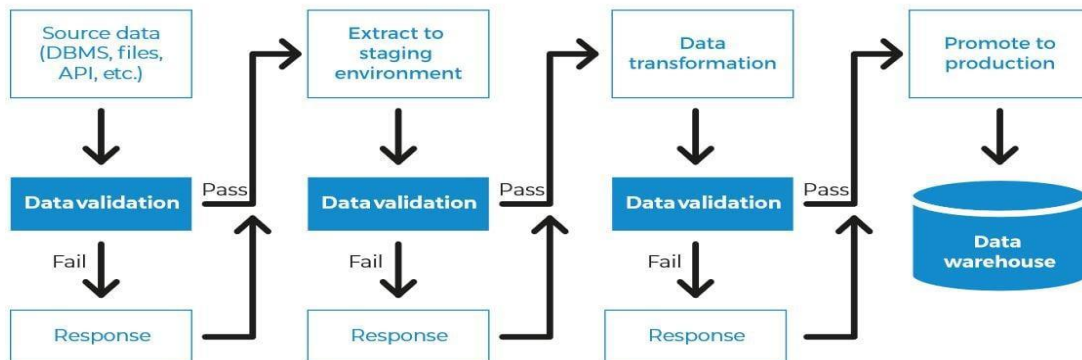
- **Purpose:** Ensure the dashboard works across different environments
- **Steps:**
 - Test in Power BI Desktop and Power BI Service
 - Check display on different screen sizes
- **Outcome:** Consistent performance

6. Security Validation

- **Purpose:** Ensure safe and secure data handling
- **Steps:**
 - Verify access permissions and sharing settings
 - Implement Row-Level Security (RLS) if required
 - Ensure sensitive data is protected
- **Outcome:** Secure dashboard environment

7. Regression Validation

- **Purpose:** Ensure updates do not affect existing functionality
- **Steps:**
 - Re-test dashboard after data refresh or changes
 - Verify that KPIs and visuals remain accurate
- **Outcome:** Stable and reliable system



6.3 DATABASE

The database plays a crucial role in storing, managing, and retrieving data efficiently for analysis and visualization in the Supply Chain & Logistics Dashboard. In this project, the dataset acts as the primary source of information used to generate insights and build interactive dashboards in Power BI.

For this project, data is sourced from structured files such as Excel/CSV, which contain detailed information related to supply chain operations. The dataset includes key attributes such as procurement cost, order quantity, shipment details, delivery status, inventory levels, production data, and sales performance.

The database structure is designed in a tabular format, where each row represents a transaction or record (e.g., order, shipment, or production entry) and each column represents a specific attribute. This structured design allows efficient data handling and analysis using Power BI.

Data preprocessing is an essential step in managing the database. It involves:

- Handling missing values
- Removing duplicate records
- Standardizing formats (dates, categories, units)
- Ensuring data consistency and accuracy

These steps improve the reliability and quality of the data used in the dashboard.

In Power BI, the data is further organized using data modeling techniques. Relationships are created between different tables, and calculated columns and measures are developed using DAX to derive KPIs such as revenue, profit, delivery performance, and inventory metrics.

Although this project uses a static dataset, in real-world applications, databases such as MySQL, SQL Server, or cloud-based systems are used for handling large-scale data. These databases can be integrated with Power BI to enable real-time data updates and dynamic reporting.

Overall, the database component ensures that data is well-structured, accurate, and ready for analysis, forming the backbone of the dashboard.

ABOUT THE DATA

The dataset used in this project represents Supply Chain and Logistics Data, which includes information related to procurement, production, inventory, shipments, and sales activities.

The primary purpose of this dataset is to analyze operational performance and support decision-making using Key Performance Indicators (KPIs).

These KPIs include:

- Revenue and Profit
- Shipment Performance (On-time delivery, delays)
- Inventory Metrics (stock levels, turnover)
- Production Efficiency and Quality

The dataset also provides insights into operational efficiency, cost management, and supply chain performance. It helps in identifying delays, reducing costs, improving delivery performance, and optimizing inventory levels.

This dataset is highly suitable for building an interactive Power BI dashboard to monitor supply chain operations, track KPIs, and generate meaningful insights.

USE OF EACH TABLE

The Power BI dashboard is built using a combination of fact tables and dimension tables. Each table plays a specific role in data storage, analysis, and KPI generation.

1. dim_customer

This table contains customer-related information such as customer ID, name, region, or segment. Helps analyze sales and orders based on customer segments and regions.

customer_id ▾	customer_key ▾	customer_name ▾	country ▾	channel_type ▾	size ▾
1	CUST-US-001	Best Buy	USA	Retailer	Large
2	CUST-US-002	Amazon	USA	Online	Large
3	CUST-US-003	Walmart	USA	Retailer	Large
4	CUST-US-004	Target	USA	Retailer	Medium
5	CUST-EU-001	MediaMarkt	Germany	Retailer	Large
6	CUST-EU-002	Carrefour	France	Retailer	Large
7	CUST-UK-001	Currys	UK	Retailer	Medium
8	CUST-JP-001	Yodobashi Camera	Japan	Retailer	Large
9	CUST-CN-001	JD.com	China	Online	Large
10	CUST-IN-001	Flipkart	India	Online	Large
11	CUST-IN-002	Reliance Digital	India	Retailer	Medium
12	CUST-KR-001	Samsung Store	South Korea	Direct	Large

2. dim_date

This table stores date-related attributes such as day, month, year, and time hierarchy. Enables time-based analysis like monthly sales trends, yearly growth, and time comparisons.

	✕ ✓						
	date_key ▼	date ▼	year ▼	quarter ▼	month ▼	month_name ▼	
	182	Saturday, July 1, 2023	2023	3	7	Jul	
	183	Sunday, July 2, 2023	2023	3	7	Jul	
	184	Monday, July 3, 2023	2023	3	7	Jul	
	185	Tuesday, July 4, 2023	2023	3	7	Jul	
	186	Wednesday, July 5, 2023	2023	3	7	Jul	
	187	Thursday, July 6, 2023	2023	3	7	Jul	
	188	Friday, July 7, 2023	2023	3	7	Jul	
	189	Saturday, July 8, 2023	2023	3	7	Jul	
	190	Sunday, July 9, 2023	2023	3	7	Jul	
	191	Monday, July 10, 2023	2023	3	7	Jul	
	192	Tuesday, July 11, 2023	2023	3	7	Jul	
	193	Wednesday, July 12, 2023	2023	3	7	Jul	
	194	Thursday, July 13, 2023	2023	3	7	Jul	
	195	Friday, July 14, 2023	2023	3	7	Jul	
	196	Saturday, July 15, 2023	2023	3	7	Jul	
	197	Sunday, July 16, 2023	2023	3	7	Jul	
	198	Monday, July 17, 2023	2023	3	7	Jul	
	199	Tuesday, July 18, 2023	2023	3	7	Jul	
	200	Wednesday, July 19, 2023	2023	3	7	Jul	
	201	Thursday, July 20, 2023	2023	3	7	Jul	
	202	Friday, July 21, 2023	2023	3	7	Jul	
	203	Saturday, July 22, 2023	2023	3	7	Jul	
	204	Sunday, July 23, 2023	2023	3	7	Jul	
	205	Monday, July 24, 2023	2023	3	7	Jul	

Table: dim_date (731 rows)

3. dim_facility

This table includes information about warehouses or facilities.

Helps track inventory, shipments, and operations across different locations.

	facility_id	facility_key	facility_name	country	city	facility_type	specialization
	1	FAC-KR-GUM-01	Gumi Plant 1	South Korea	Gumi	Manufacturing	Smartphone
	2	FAC-KR-GUM-02	Gumi Plant 2	South Korea	Gumi	Manufacturing	Component
	3	FAC-VN-TNG-01	Thai Nguyen Plant	Vietnam	Thai Nguyen	Manufacturing	Smartphone
	4	FAC-VN-BGG-01	Bac Giang Plant	Vietnam	Bac Giang	Manufacturing	Smartphone
	5	FAC-IN-NOI-01	Noida Plant	India	Noida	Manufacturing	Smartphone
	6	FAC-BR-CMP-01	Campinas Plant	Brazil	Campinas	Manufacturing	Electronics
	7	FAC-KR-TAN-01	Tangjeong Plant	South Korea	Asan	Manufacturing	Display
	8	FAC-KR-SWN-01	Suwon R&D	South Korea	Suwon	R&D	Technology
	9	WH-US-NJ-01	New Jersey DC	USA	Cranbury	Warehouse	Distribution
	10	WH-EU-PL-01	Poland DC	Poland	Warsaw	Warehouse	Distribution
	11	WH-CN-SH-01	Shanghai DC	China	Shanghai	Warehouse	Distribution
	12	WH-SG-SG-01	Singapore DC	Singapore	Singapore	Warehouse	Distribution

4. dim_product

This table contains product details such as product name, category, and type.

Used to analyze product-wise sales, inventory, and performance.

	product_id	product_key	product_name	category	product_line	spec	color	unit_price	Img
	1	SMRT-GS24U-512-BLK	Galaxy S24 Ultra	Smartphone	Galaxy S	512GB	Black	1199.99	https://i.ibb.co/9Hv19S22/s24-Ultra.webp
	2	SMRT-GS24U-256-GRY	Galaxy S24 Ultra	Smartphone	Galaxy S	256GB	Gray	1099.99	https://i.ibb.co/9Hv19S22/s24-Ultra.webp
	3	SMRT-GS24-256-VLT	Galaxy S24	Smartphone	Galaxy S	256GB	Violet	799.99	https://i.ibb.co/gMJgXrhG/s24.webp
	4	SMRT-GS23-128-CRM	Galaxy S23	Smartphone	Galaxy S	128GB	Cream	699.99	https://i.ibb.co/tPH2nTkP/s23.webp
	5	SMRT-GZ5F-512-BLK	Galaxy Z Fold5	Smartphone	Galaxy Z	512GB	Black	1799.99	https://i.ibb.co/tPnpz3yp/Fold5.webp
	6	SMRT-GZ5F-256-BLU	Galaxy Z Fold5	Smartphone	Galaxy Z	256GB	Blue	1599.99	https://i.ibb.co/kgXHN1Nh/fold-5-2.webp
	7	SMRT-GA54-128-BLK	Galaxy A54	Smartphone	Galaxy A	128GB	Black	449.99	https://i.ibb.co/HDYmTQ8b/a54.webp
	8	SMRT-GA34-64-WHT	Galaxy A34	Smartphone	Galaxy A	64GB	White	349.99	https://i.ibb.co/F4brOpKx/a34.webp
	9	TBLT-GTS9U-512-GRY	Galaxy Tab S9 Ultra	Tablet	Tab S	512GB	Gray	1199.99	https://i.ibb.co/v6MTK3p6/s9.webp
	10	TBLT-GTS9-256-BLK	Galaxy Tab S9	Tablet	Tab S	256GB	Black	799.99	https://i.ibb.co/Xf1xx78G/tab-s9-ultra.webp
	11	TBLT-GTA9-128-SLV	Galaxy Tab A9	Tablet	Tab A	128GB	Silver	299.99	https://i.ibb.co/p67NdXxW/tab-a9.webp
	12	TV-QLED85-85IN-BLK	QLED 85" 4K	Television	QLED	85 inch	Black	2799.99	https://i.ibb.co/MDMdxTmG/q-85.webp
	13	TV-QLED75-75IN-BLK	QLED 75" 4K	Television	QLED	75 inch	Black	1999.99	https://i.ibb.co/KzKrmT0s/q-75.webp
	14	TV-OLED65-65IN-BLK	OLED 65" 4K	Television	OLED	65 inch	Black	2499.99	https://i.ibb.co/mw7XszY/q-65.webp
	15	TV-CRYS55-55IN-BLK	Crystal UHD 55"	Television	Crystal	55 inch	Black	699.99	https://i.ibb.co/nsB3JCL/Crystal-UHD-55.webp
	16	APPL-RF28-ADR-SS	Family Hub Refrigerator	Appliance	Refrigerator	28 cu ft	Stainless	3299.99	https://i.ibb.co/N2MsQ6ZX/Family-Hub-Refrigerator.webp
	17	APPL-WM50-FRT-WHT	Front Load Washer	Appliance	Washer	5.0 cu ft	White	1099.99	https://i.ibb.co/xqKy25zv/Front-Load-Washer.webp
	18	APPL-DW24-BLT-BLK	Built-in Dishwasher	Appliance	Dishwasher	24 inch	Black	799.99	https://i.ibb.co/hJjm4wW/Built-in-Dishwasher.webp
	19	WEAR-GW6-44MM-BLK	Galaxy Watch6	Wearable	Watch	44mm	Black	329.99	https://i.ibb.co/KzjsL0KT/Galaxy-Watch6.webp
	20	WEAR-GB2P-GRY	Galaxy Buds2 Pro	Wearable	Earbuds	Standard	Gray	229.99	https://i.ibb.co/j9vPKWS2/Galaxy-Buds2-Pro.webp

5. dim_suppliers

This table includes supplier information such as supplier ID, name, and details. Helps evaluate supplier performance and procurement efficiency.

supplier_id	supplier_key	supplier_name	country	city	specialty	tier
1	SUP-KR-001	Samsung Electronics Co.	South Korea	Seoul	Components	Tier 1
2	SUP-KR-002	SK Hynix	South Korea	Icheon	Memory Chips	Tier 1
3	SUP-TW-001	TSMC	Taiwan	Hsinchu	Processors	Tier 1
4	SUP-JP-001	Sony Corporation	Japan	Tokyo	Camera Sensors	Tier 1
5	SUP-CN-001	BOE Technology	China	Beijing	Display Panels	Tier 1
6	SUP-KR-003	LG Display	South Korea	Seoul	OLED Panels	Tier 1
7	SUP-US-001	Qualcomm	USA	San Diego	Chipsets	Tier 1
8	SUP-KR-004	Samsung SDI	South Korea	Yongin	Batteries	Tier 1
9	SUP-CN-002	Foxconn	China	Shenzhen	Assembly	Tier 2
10	SUP-VN-001	Vina Samsung	Vietnam	Thai Nguyen	Assembly	Tier 2
11	SUP-IN-001	Samsung India	India	Noida	Assembly	Tier 2
12	SUP-JP-002	Murata Manufacturing	Japan	Kyoto	Capacitors	Tier 2
13	SUP-KR-005	Samsung Electro-Mechanics	South Korea	Suwon	PCB Boards	Tier 2

6. fact_sales

This is the primary table containing sales transaction data such as revenue, quantity, and orders. Used to calculate KPIs like Total Revenue, Profit, and Sales Trends.

	sales_id	date_key	product_id	customer_id	quantity_sold	unit_price	discount_pct	discount_amount	revenue	cost	profit
	48	549	3	9	455	799.99	0	0	363995.45	272949.34	91046.11
	49	657	3	11	414	799.99	0	0	331195.86	250424.99	80770.87
	96	105	3	7	631	799.99	0	0	504793.69	372238.03	132555.66
	160	416	3	12	355	799.99	0	0	283996.45	215754.33	68242.12
	186	722	3	1	262	799.99	0	0	209597.38	175897.97	33699.41
	242	389	3	5	650	799.99	0	0	519993.5	428096.96	91896.54
	391	181	3	4	448	799.99	0	0	358395.52	269001.34	89394.18
	426	127	3	7	142	799.99	0	0	113598.58	90364.95	23233.63
	472	610	3	10	792	799.99	0	0	633592.08	516059.63	117532.45
	513	149	3	4	21	799.99	0	0	16799.79	13671.17	3128.62
	543	672	3	5	196	799.99	0	0	156798.04	122236.09	34561.95
	624	689	3	4	160	799.99	0	0	127998.4	94757.47	33240.93
	634	182	3	2	106	799.99	0	0	84798.94	71416.56	13382.38
	683	494	3	11	235	799.99	0	0	187997.65	156108.11	31889.54
	695	497	3	12	633	799.99	0	0	506393.67	359780	146613.67
	725	464	3	3	83	799.99	0	0	66399.17	55731.32	10667.85
	729	667	3	3	794	799.99	0	0	635192.06	512022.67	123169.39
	782	345	3	8	769	799.99	0	0	615192.31	503608.48	111583.83
	827	604	3	1	46	799.99	0	0	36799.54	30581.57	6217.97
	829	317	3	2	778	799.99	0	0	622392.22	518188.27	104203.95
	834	83	3	2	340	799.99	0	0	271996.6	195858.43	76138.17
	835	534	3	9	327	799.99	0	0	261596.73	210587.73	51009
	869	455	3	3	526	799.99	0	0	420794.74	295823.05	124971.69
	952	708	3	10	812	799.99	0	0	649591.88	552098.22	97493.66

Table: fact_sales (8,000 rows)

7. fact_shipment

This table stores shipment details such as delivery status, shipping cost, and shipment count.

Use: Used to measure delivery performance, on-time delivery rate, and logistics efficiency.

supplier_id	supplier_key	supplier_name	country	city	specialty	tier
1	SUP-KR-001	Samsung Electronics Co.	South Korea	Seoul	Components	Tier 1
2	SUP-KR-002	SK Hynix	South Korea	Icheon	Memory Chips	Tier 1
3	SUP-TW-001	TSMC	Taiwan	Hsinchu	Processors	Tier 1
4	SUP-JP-001	Sony Corporation	Japan	Tokyo	Camera Sensors	Tier 1
5	SUP-CN-001	BOE Technology	China	Beijing	Display Panels	Tier 1
6	SUP-KR-003	LG Display	South Korea	Seoul	OLED Panels	Tier 1
7	SUP-US-001	Qualcomm	USA	San Diego	Chipsets	Tier 1
8	SUP-KR-004	Samsung SDI	South Korea	Yongin	Batteries	Tier 1
9	SUP-CN-002	Foxconn	China	Shenzhen	Assembly	Tier 2
10	SUP-VN-001	Vina Samsung	Vietnam	Thai Nguyen	Assembly	Tier 2
11	SUP-IN-001	Samsung India	India	Noida	Assembly	Tier 2
12	SUP-JP-002	Murata Manufacturing	Japan	Kyoto	Capacitors	Tier 2
13	SUP-KR-005	Samsung Electro-Mechanics	South Korea	Suwon	PCB Boards	Tier 2

8. fact_inventory

This table contains inventory-related data such as stock levels and inventory value.

Helps calculate stock levels, inventory turnover, and stock availability.

inventory_id	date_key	product_id	facility_id	stock_level	safety_stock_level	reorder_point
4	454	3	10	36693	8449	9413
21	707	11	10	5005	1386	1543
37	523	4	10	11371	1731	3886
38	382	13	10	12954	2288	4529
57	333	7	10	14909	4341	4017
69	430	16	10	19093	4042	7600
72	81	8	10	21436	5203	6293
73	653	11	10	8951	2288	2324
77	202	17	10	24942	7046	7556
82	684	14	10	3881	812	1156
84	597	2	10	45928	9627	12158
98	677	19	10	2408	535	697
119	687	8	10	27337	5596	7490
120	644	6	10	20868	4948	6920
122	530	19	10	46907	12455	15063
123	187	19	10	25277	6100	8599
135	731	6	10	14701	3963	4462
137	199	12	10	42627	9015	13754
154	169	4	10	19091	2888	7373
177	200	12	10	31052	6119	11986
205	437	7	10	2416	662	795
238	449	2	10	38081	8579	12078
248	664	17	10	29679	4984	11443
265	75	1	10	13813	8882	17063

Table: fact_inventory (4,000 rows)

9. fact_production

This table includes production data such as units produced, defective units, and efficiency metrics.

Use: Used to analyze production performance, defect rate, and quality yield.

	production_id	date_key	product_id	facility_id	quantity_produced	defective_units	defect_rate_pct
	7	349	9	2	5527	132	2.4
	18	656	18	2	1628	34	2.12
	21	581	11	2	12738	165	1.3
	22	659	15	2	2669	11	0.44
	36	372	10	2	2949	20	0.68
	38	546	5	2	48237	598	1.24
	44	725	2	2	9417	213	2.27
	51	112	8	2	17465	242	1.39
	52	144	14	2	2781	33	1.21
	55	498	16	2	1942	44	2.27
	58	570	16	2	1077	8	0.81
	73	680	4	2	22332	84	0.38
	74	110	18	2	1415	11	0.78
	75	216	11	2	13263	214	1.62
	82	597	18	2	2060	8	0.41
	119	562	15	2	4388	51	1.18
	123	554	3	2	14884	98	0.66
	129	560	20	2	8999	56	0.63
	142	263	12	2	1438	28	1.97
	155	507	11	2	9988	60	0.61
	164	314	9	2	3976	72	1.83
	166	707	6	2	19180	358	1.87
	175	703	2	2	14777	302	2.05
	176	312	3	2	12762	183	1.44

Table: fact_production (5,000 rows)

10. fact_procurement

This table contains procurement-related data such as purchase cost and supplier transactions. Helps track procurement cost and cost efficiency.

	procurement_id	order_date_key	product_id	supplier_id	order_quantity	unit_cost	total_cost	lead_time_days
	1	392	17	7	14004	345.33	4836001.32	27
	17	486	6	7	81874	16.19	1325540.06	25
	21	360	4	7	5856	70.05	410212.8	20
	23	441	18	7	93159	146.34	13632888.06	40
	29	542	7	7	80682	481.16	38820951.12	22
	42	85	18	7	63918	499.99	31958360.82	38
	50	721	9	7	57615	223.6	12882714	17
	66	265	10	7	96080	27.77	2668141.6	33
	67	148	15	7	29673	426.8	12664436.4	33
	70	359	18	7	11888	248.13	2949769.44	40
	72	685	6	7	60383	493.5	29799010.5	39
	96	478	18	7	6975	391.43	2730224.25	20
	103	480	4	7	14865	216.24	3214407.6	21
	120	374	18	7	92525	148.95	13781598.75	29
	129	379	9	7	61630	236.94	14602612.2	16
	137	460	14	7	33856	72.37	2450158.72	42
	139	214	18	7	63375	211.7	13416487.5	22
	151	380	16	7	80527	64.74	5213317.98	45
	154	180	4	7	76836	8.89	683072.04	25
	181	302	8	7	22392	298.09	6674831.28	22
	184	342	20	7	42032	454.3	19095137.6	22
	214	621	14	7	23313	416.58	9711729.54	42
	220	607	8	7	1758	348.94	613436.52	40
	222	447	5	7	37881	112.1	4244341.1	36

Table: fact_procurement (3,000 rows)

11. Samsung / Table / Table (2)

These appear to be additional or temporary tables used for testing or specific calculations.
May support custom visuals, intermediate calculations, or filtered datasets.

Name ▾	Extension ▾	Date accessed ▾	Date modified ▾	Date created ▾	Folder Path ▾
dim_date.csv	.csv	1/13/2026 2:48:23 AM	1/13/2026 2:47:40 AM	1/13/2026 2:48:23 AM	C:\Users\faisa\OneDrive\Documents\Samsung\
dim_distributors.csv	.csv	1/13/2026 2:48:23 AM	1/13/2026 2:47:41 AM	1/13/2026 2:48:23 AM	C:\Users\faisa\OneDrive\Documents\Samsung\
dim_plants.csv	.csv	1/13/2026 2:48:23 AM	1/13/2026 2:47:43 AM	1/13/2026 2:48:23 AM	C:\Users\faisa\OneDrive\Documents\Samsung\
dim_products.csv	.csv	1/13/2026 2:48:39 AM	1/13/2026 2:47:45 AM	1/13/2026 2:48:23 AM	C:\Users\faisa\OneDrive\Documents\Samsung\
dim_suppliers.csv	.csv	1/13/2026 2:48:23 AM	1/13/2026 2:47:48 AM	1/13/2026 2:48:23 AM	C:\Users\faisa\OneDrive\Documents\Samsung\
dim_warehouses.csv	.csv	1/13/2026 2:49:04 AM	1/13/2026 2:47:49 AM	1/13/2026 2:48:23 AM	C:\Users\faisa\OneDrive\Documents\Samsung\
fact_inventory_snapshot.csv	.csv	1/13/2026 2:48:23 AM	1/13/2026 2:47:52 AM	1/13/2026 2:48:23 AM	C:\Users\faisa\OneDrive\Documents\Samsung\
fact_manufacturing.csv	.csv	1/13/2026 2:48:23 AM	1/13/2026 2:47:56 AM	1/13/2026 2:48:23 AM	C:\Users\faisa\OneDrive\Documents\Samsung\
fact_orders.csv	.csv	1/13/2026 2:48:23 AM	1/13/2026 2:47:59 AM	1/13/2026 2:48:23 AM	C:\Users\faisa\OneDrive\Documents\Samsung\

7. Dashboard Components

Dashboard Components

The Power BI dashboard is designed using KPI cards that represent key performance metrics of the Supply Chain & Logistics system. These KPIs are directly derived from the dataset and visualized to support data-driven decision-making.

7.1 Financial KPIs

1. Total Revenue

- **Value:** 4.50 bn
- **Formula:** $\text{SUM}(\text{Sales Amount})$
- **Description:** Represents the total income generated from all sales transactions.

2. Total Profit

- **Value:** 1.01 bn
- **Formula:** $\text{Revenue} - \text{Cost}$
- **Description:** Shows the net earnings after deducting procurement and operational costs.

3. Profit Margin (%)

- **Value:** 22.5%
- **Formula:** $(\text{Profit} \div \text{Revenue}) \times 100$
- **Description:** Indicates overall profitability efficiency.

4. Cost Ratio (%)

- **Value:** 864.83
- **Formula:** $(\text{Total Cost} \div \text{Revenue})$
- **Description:** Measures cost efficiency relative to revenue.

5. Discount Ratio (%)

- **Value:** 7.17%
- **Formula:** $(\text{Discount} \div \text{Total Sales}) \times 100$
- **Description:** Shows the impact of discounts on revenue.

7.2 Procurement & Order KPIs

6. Procurement Cost

- **Value:** 38.89 bn
- **Formula:** SUM(Procurement Cost)
- **Description:** Total cost incurred in purchasing goods.

7. Total Order Quantity

- **Value:** 152 M
- **Formula:** SUM(Order Quantity)
- **Description:** Total number of units ordered.

8. Average Unit Cost

- **Value:** 255.10
- **Formula:** Procurement Cost $\div$ Quantity
- **Description:** Average cost per unit of product.

7.3 Logistics & Delivery KPIs

9. Total Shipments

- **Value:** 6K
- **Formula:** COUNT(Shipment ID)
- **Description:** Total number of shipments processed.

10. On-Time Delivery (%)

- **Value:** 50.37%
- **Formula:** (On-Time Deliveries $\div$ Total Shipments) $\times$ 100
- **Description:** Measures delivery punctuality.

11. Delivery Reliability (%)

- **Value:** 70.47%
- **Formula:** (Successful Deliveries $\div$ Total Deliveries) $\times$ 100
- **Description:** Indicates reliability of delivery operations.

12. In-Transit Shipments

- **Value:** 1K
- **Formula:** COUNT(Status = "In Transit")

- **Description:** Shipments currently in progress.

13. Total Shipping Cost

- **Value:** 41.43 M
- **Formula:** SUM(Shipping Cost)
- **Description:** Total logistics cost for deliveries.

14. Delivery Score

- **Value:** 31.96
- **Formula:** Composite KPI (based on delivery metrics)
- **Description:** Overall delivery performance indicator.

7.4 Production KPIs

15. Total Units Produced

- **Value:** 68 M
- **Formula:** SUM(Production Quantity)

16. Defective Units

- **Value:** 886 K
- **Formula:** SUM(Defective Units)

17. Defect Rate (%)

- **Value:** 1.30%
- **Formula:** $(\text{Defective Units} \div \text{Total Units}) \times 100$

18. Quality Yield (%)

- **Value:** 98.70%
- **Formula:** $(\text{Good Units} \div \text{Total Units}) \times 100$

19. Production Efficiency

- **Value:** 98.70%
- **Description:** Measures efficiency of production process.

7.5 Inventory KPIs

20. Current Stock Level

- **Value:** 100 M
- **Formula:** SUM(Stock Quantity)

21. Inventory Turnover

- **Value:** 4.03%
- **Formula:** Cost of Goods Sold ÷ Average Inventory

22. Days of Inventory

- **Value:** 9.07 K
- **Formula:** (Inventory ÷ Daily Usage)

23. Inventory Value

- **Value:** 90.16 bn
- **Formula:** SUM(Inventory Value)

7.6 Sales & Shipment KPIs

24. Quantity Shipped

- **Value:** 15 M
- **Formula:** SUM(Shipped Quantity)

25. Delivered Quantity

- **Value:** 8 M
- **Formula:** SUM(Delivered Quantity)

8. OUTPUT

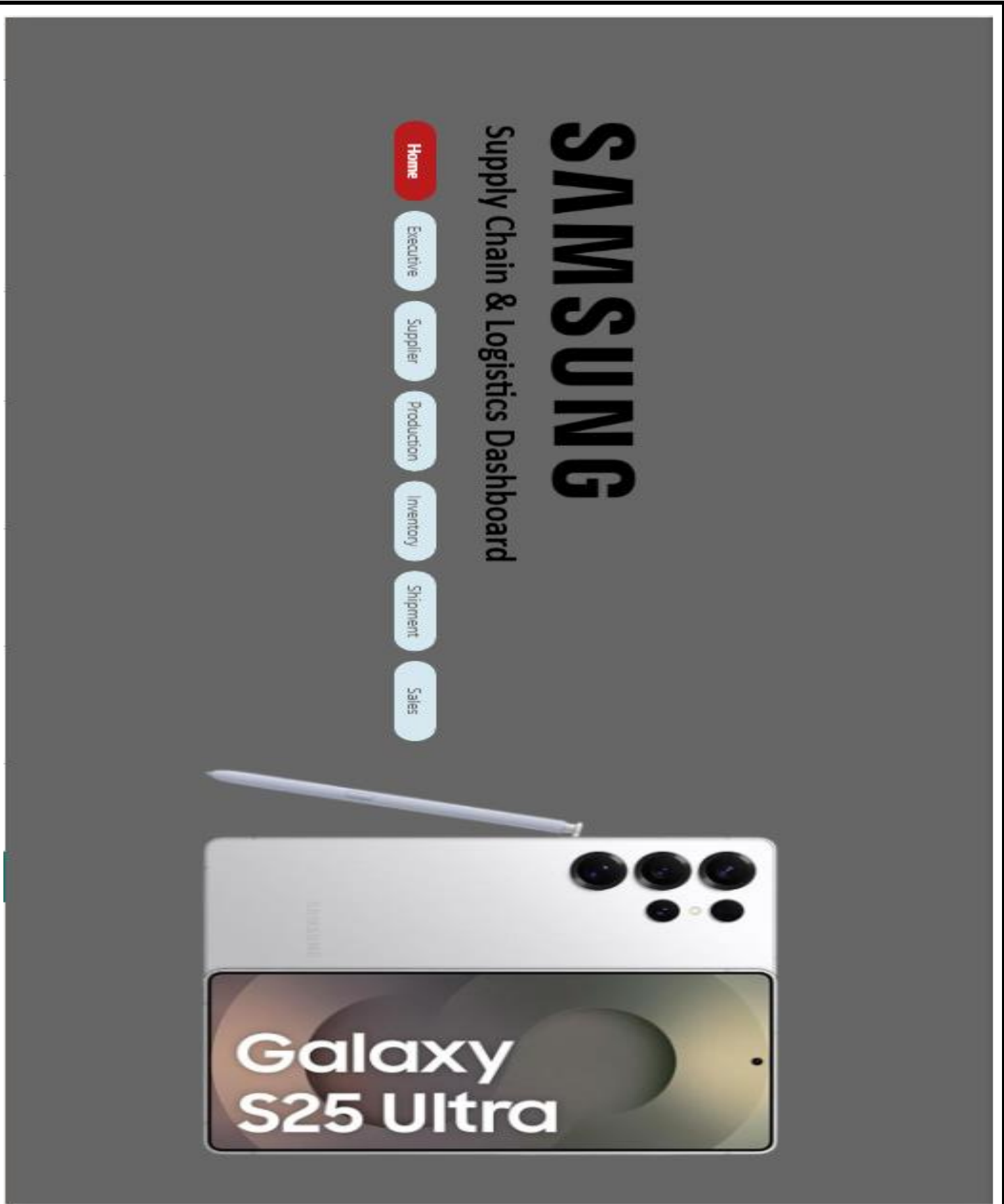


Figure 1.0

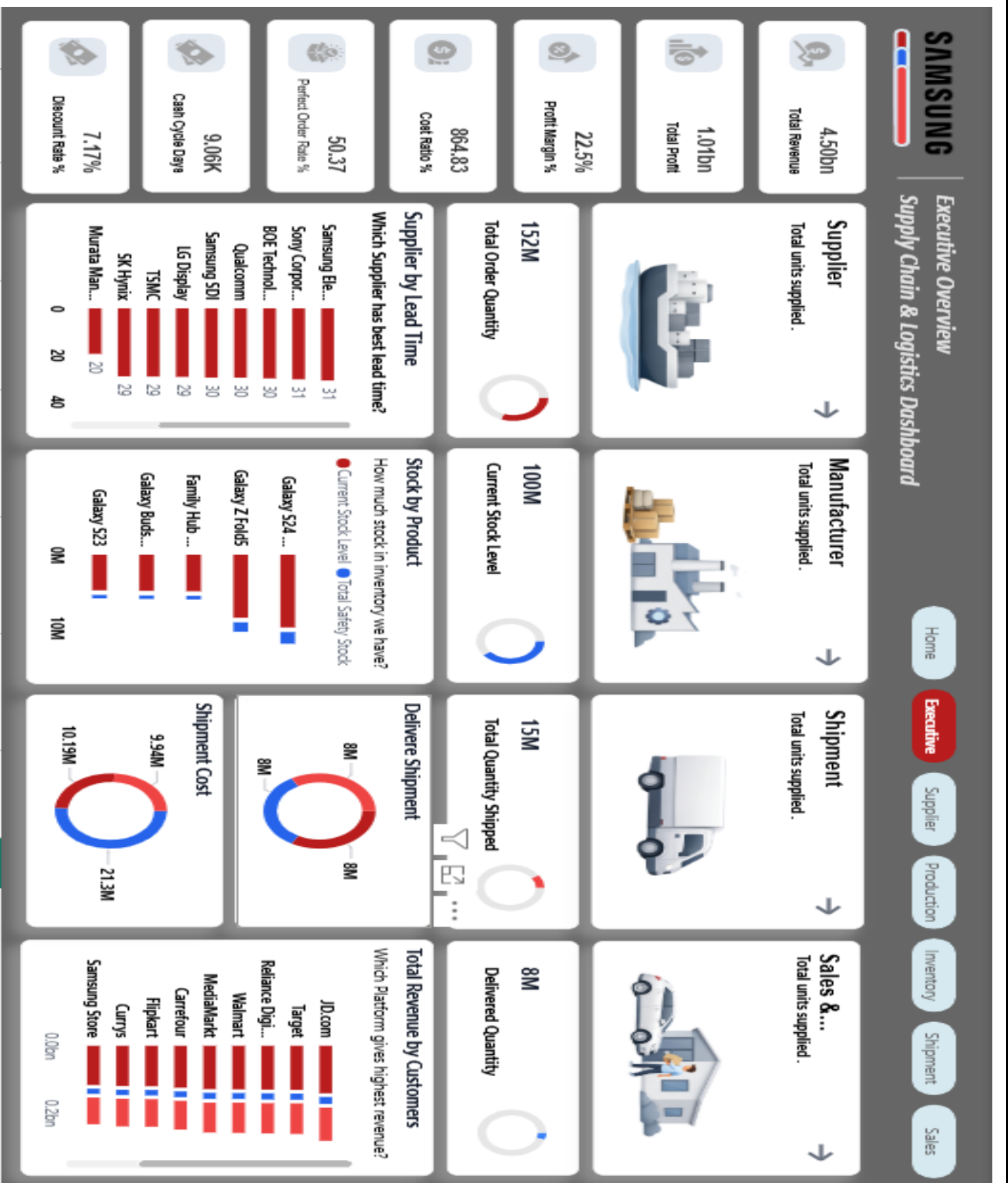
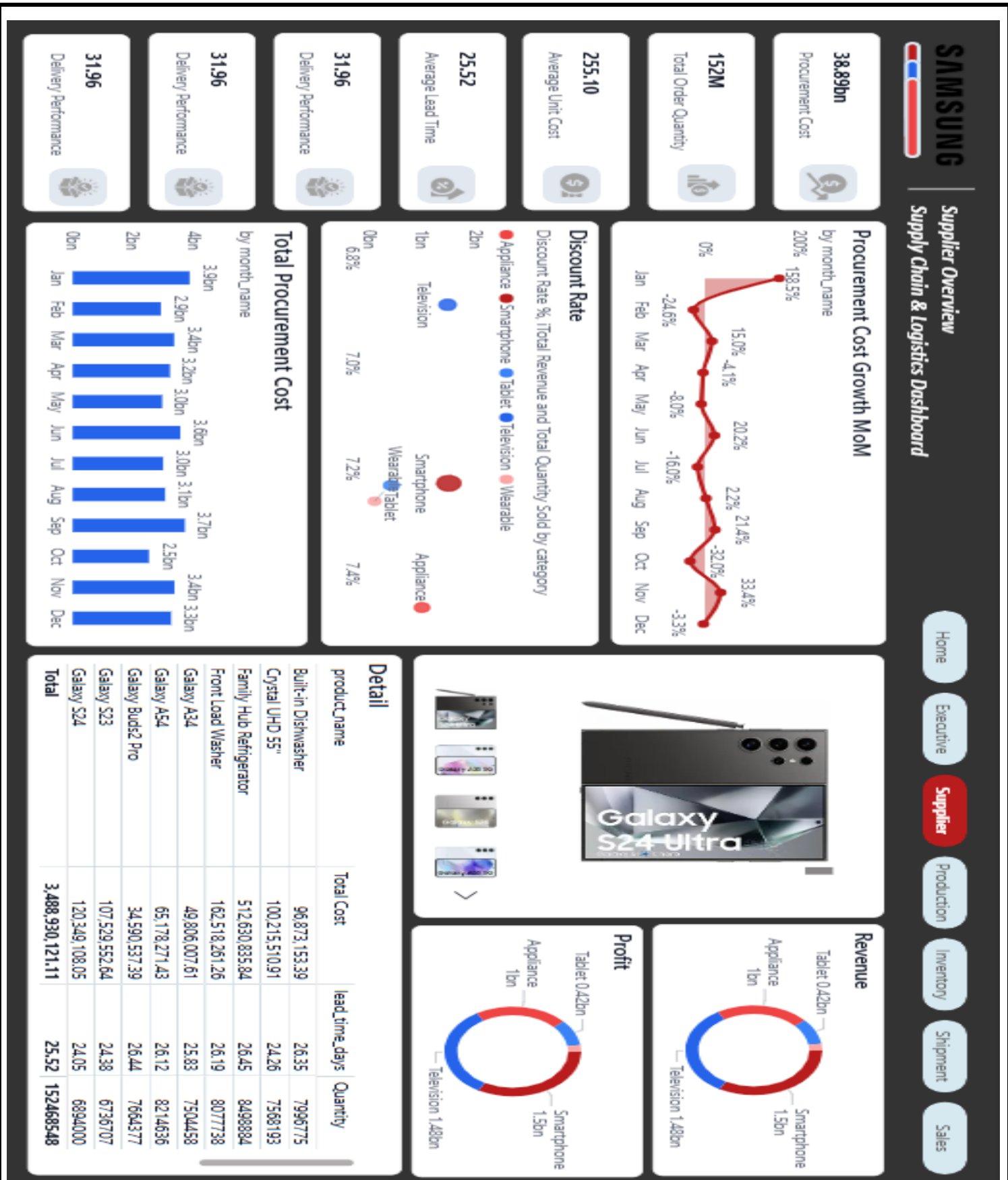


Figure 1.1



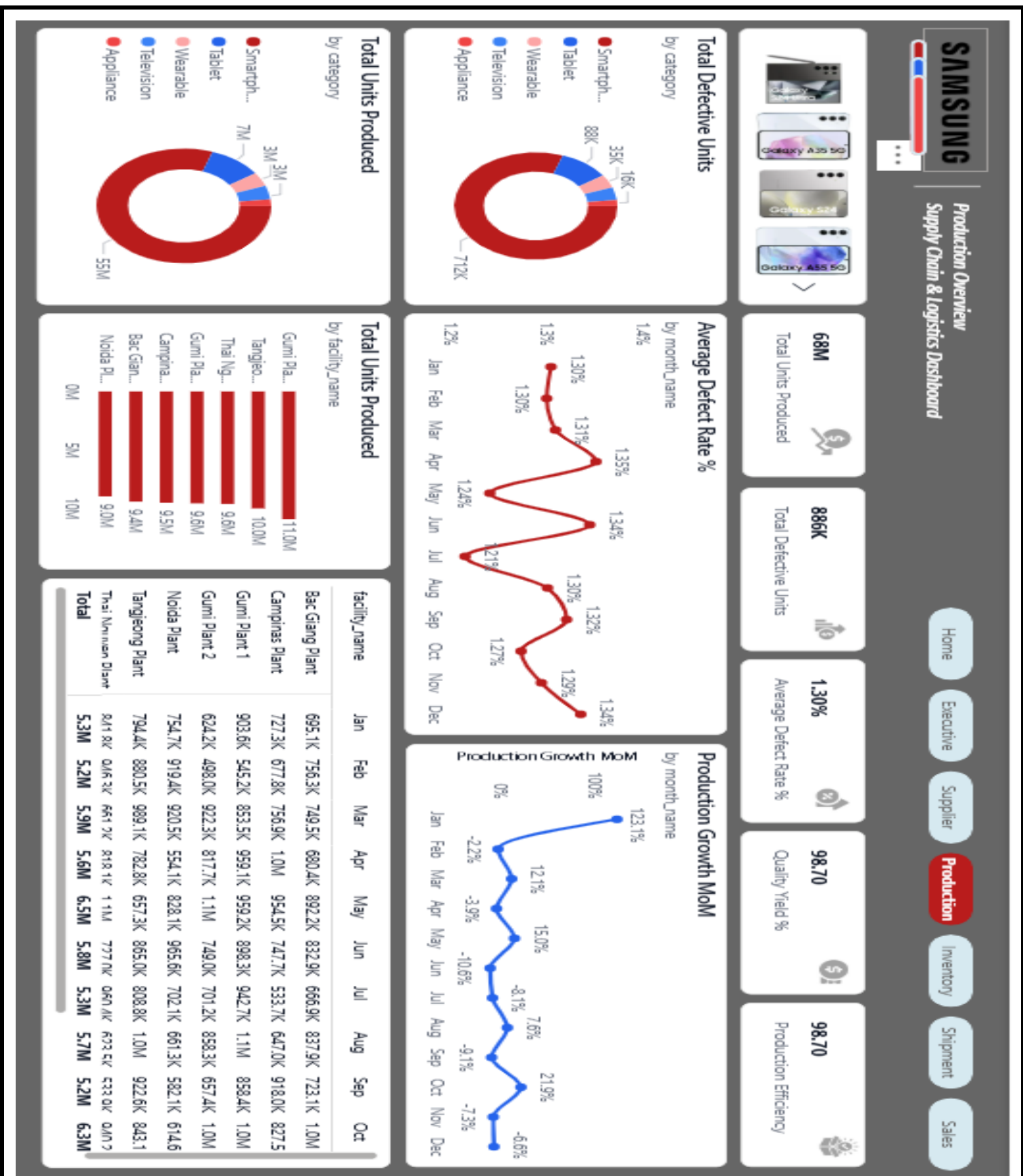


Figure 1.3

Inventory Performance Supply Chain & Logistics Dashboard

Sales



by product_name

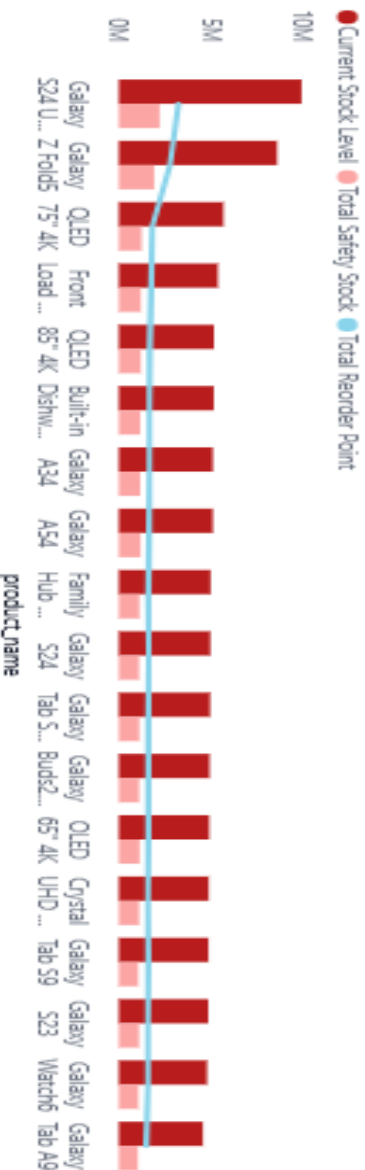


Figure 1.4

Shipment Performance
Supply Chain & Logistics Dashboard

Sales

Total Shipments

On-Time Delivery %

On-Time Delivery R...

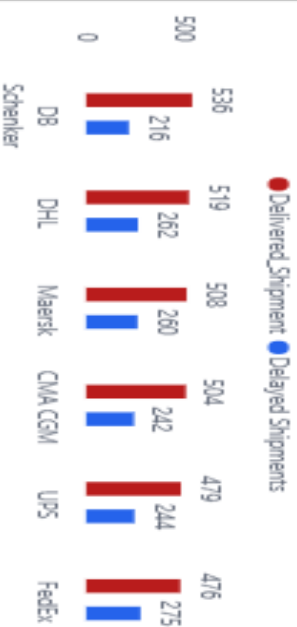
In Transit Shipments

Total Shipping Cost

Delivery Score

Quantity Shipped

by carrier



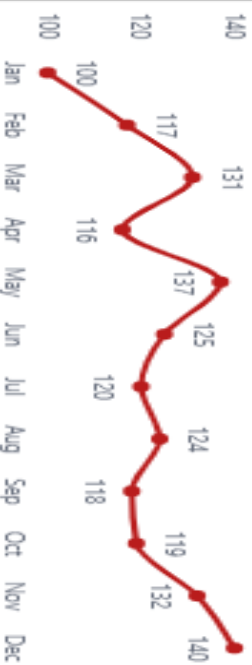
Delivery and Delayed



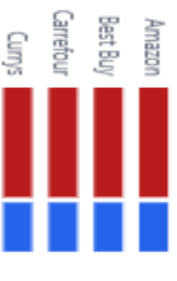
Delivery and Delayed



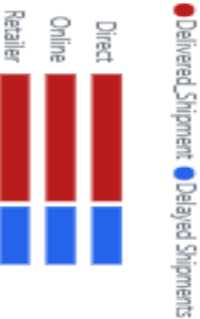
by month_name



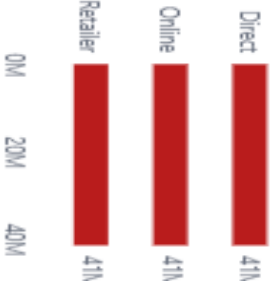
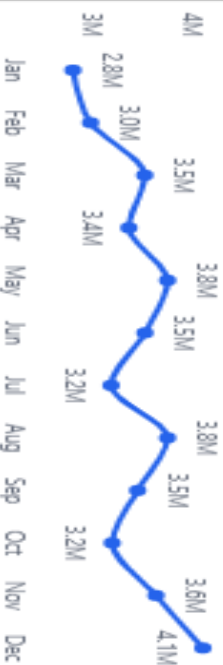
● Delivered Shipment ● Delayed Shipments



By Channel



by month_name



by Customer

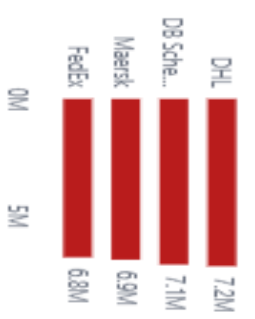


Figure 1.5

SAMSUNG

Sales Performance
Supply Chain & Logistics Dashboard

Home

Executive

Supplier

Production

Inventory

Shipment

Sales

6K

Total Shipments



50.37%

On-Time Delivery %



70.47%

On-Time Delivery R...



1K

In Transit Shipments



41.43M

Total Shipping Cost



31.96

Delivery Score



15M

Quantity Shipped



Image URL Total Revenue Profit Margin %

Image URL	Total Revenue	Profit Margin %
Image URL	4,502.12M	22.5%
Image URL	4,502.12M	22.5%
Image URL	4,502.12M	22.5%
Image URL	4,502.12M	22.5%
Image URL	4,502.12M	22.5%
Image URL	4,502.12M	22.5%
Image URL	4,502.12M	22.5%

Total Revenue, Total Profit and Revenue Growth YoY

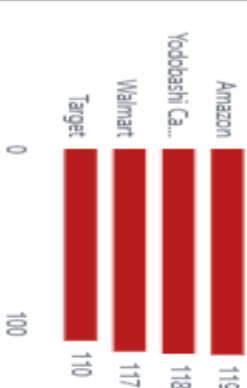
by month_name

Total Revenue Total Profit Revenue Growth YoY



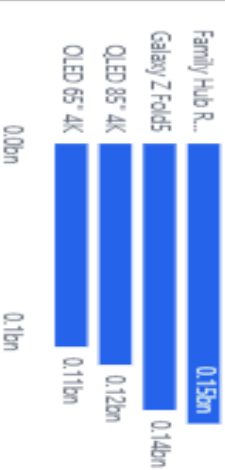
Revenue Growth YoY

by customer_name



Total Profit

by product_name



Discount Rate %, Total Revenue and Total Quantity Sold

by category

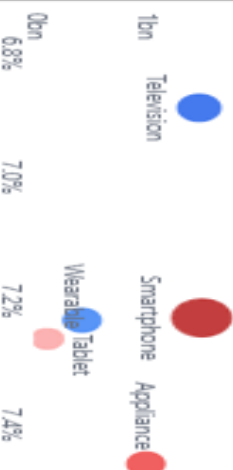


Figure 1.6

9. CONCLUSION

9. CONCLUSION

The Supply Chain & Logistics Dashboard developed using Power BI demonstrates the effective use of data analytics and visualization techniques to transform raw operational data into meaningful business insights. The project successfully integrates multiple domains of the supply chain, including procurement, production, inventory, shipment, and sales, into a single interactive platform, enabling a holistic view of organizational performance.

One of the key achievements of this project is the implementation of well-defined Key Performance Indicators (KPIs) such as Total Revenue (4.50 bn), Total Profit (1.01 bn), Profit Margin (22.5%), On-Time Delivery (50.37%), and Inventory Turnover (4.03%). These KPIs provide a clear and measurable understanding of business performance. For instance, while the organization shows strong revenue and profit generation, the relatively moderate on-time delivery percentage highlights an area for operational improvement. Similarly, production metrics such as Quality Yield (98.70%) and low Defect Rate (1.30%) indicate high efficiency and quality standards in manufacturing processes.

The dashboard's interactive features, including filters, slicers, and drill-down capabilities, allow users to explore data dynamically and gain deeper insights into specific regions, time periods, or product categories. This flexibility supports data-driven decision-making by enabling stakeholders to quickly identify trends, inefficiencies, and opportunities for optimization. The integration of Python (Pandas and Matplotlib) further enhances the analytical capability by enabling advanced data processing and customized visualizations beyond standard Power BI features.

From a system design perspective, the project adopts a structured architecture where data is collected from Excel/CSV sources, processed through Power Query and Python, and modeled within Power BI for efficient analysis. The use of DAX formulas ensures accurate calculation of

KPIs, while the optimized data model supports fast performance and scalability. Additionally, validation and testing processes ensure the reliability, accuracy, and consistency of the dashboard outputs.

This project also highlights the importance of digital transformation in modern supply chain management. By replacing manual reporting methods with an automated, real-time dashboard, the system reduces errors, improves transparency, and enhances operational efficiency. It enables better monitoring of logistics performance, cost control, inventory management, and customer satisfaction.

Despite its strengths, the project has certain limitations. The dashboard is based on a static dataset, and real-time data integration is not implemented. Features such as predictive analytics, AI-based forecasting, and automated alerts are not included but can significantly enhance the system in the future.

In conclusion, the Supply Chain & Logistics Dashboard serves as a powerful business intelligence tool that simplifies complex data into actionable insights. It not only improves decision-making but also provides a scalable foundation for future enhancements such as real-time analytics, machine learning integration, and cloud-based deployment. This project successfully demonstrates the practical application of Power BI, data analytics, and KPI-based monitoring in optimizing supply chain operations and driving business growth.

10. BIBLIOGRAPHY

- Microsoft Power BI Documentation: <https://learn.microsoft.com/en-us/power-bi/>
- Power BI Community & Tutorials: <https://community.powerbi.com/>
- W3Schools (Data Visualization Concepts): <https://www.w3schools.com/>
- Python Documentation (W3Schools): <https://www.w3schools.com/python/>
- Pandas Documentation: <https://pandas.pydata.org/docs/>
- Matplotlib Documentation: <https://matplotlib.org/stable/index.html>
- Google Scholar: <https://scholar.google.com/>
- Kaggle (Datasets & Data Analysis): <https://www.kaggle.com/>